

APPROVED



MSc Data Science
School of Computing, Engineering and Technology

Version	4.0
Course Type	Postgraduate
Award Type	Master of Science - MSc
Awards	
No Course Award Assigned	
Code	0502
SCQF Level	Level 11
QAA Subject Benchmark	Computing (2022)
SCQF Credit Points	180
ECTS Credit Points	90
Valid From	2026 September as of (September 2026)

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100992	machine learning	machine learning	100%
Total			100%

Professional, Statutory and Regulatory Body (PSRB)
No PSRBs have been recorded

	The general format of the first two semesters for full-time students comprises a mix of lectures and lab sessions. Students will have access to all materials on the University's VLE. In semester 3, students are engaged on a project module, and have a structured timetable as well as meetings with their project supervisor to discuss progress and offer support. Part-time students attend the same programmed sessions as the full-time students but over an extended duration corresponding to two 15-credit modules per semester. Full Time Students • For September entry, the sequence of modules will be: Semester 1 (taken in September), Semester 2 (taken in January), Semester 3 (project, taken in May) • For January entry, the sequence of modules will be: Semester 2 (taken in January), Semester 1 (taken in September), Semester 3 (project, taken in January)
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What the Course Involves	Part Time Students • For September entry, the sequence of modules will be: Semester 1 (taken in September), Semester 2 (taken in January), Semester 3 (taken in September), Semester 4 (taken in January), Semester 5 (project, taken in May and spanning 2 semesters) • For January entry, the sequence of modules will be: Semester 2 (taken in January), Semester 1 (taken in September), Semester 4 (taken in January), Semester 3 (taken in September), Semester 5 (project, taken in January and spanning 2 semesters) Students on the distance learning programme will not have face-to-face sessions. Instead, they will have access to lecture, tutorial and laboratory electronic (video, audio, written) materials on the University's virtual campus together with on-line support sessions. Their module schedule is the same as other part-time students as above.
Course Collaborations No collaborations listed	
Introduction to the Course	The MSc/PgD/PgC Data Science is a specialist Masters course, designed to equip Honours graduates in a discipline with significant Computing, Mathematics or Statistics content with expert knowledge of modern data management, analysis and visualisation methods, information retrieval and extraction techniques, and relational and NoSQL data stores. It is also designed to provide a foundation of research skills to help the student become an independent researcher. It provides the technical skills and research training necessary to develop sophisticated data analysis and retrieval systems applied to Big Data. The Short Industrial Placement or Industrial Internship placement modules build on knowledge and technical skills to meet challenges in the business environment. The MSc project module is of particular value to employers as evidence of the student's ability to produce well-documented high-quality work of an advance technical standard to deadline and to a high standard of presentation. At all stages of study, the course content is complemented by practical work intended to not only reinforce the learning process but also to enhance the vocational relevance of the course of study. The syllabus content also incorporates course modules intended to enhance the technical and professional skills required to manage and execute state of the art data processing systems.
	The specific aims for each stage of the course are: Postgraduate Certificate To produce practitioners who will be able to: • Demonstrate knowledge and understanding of current data management fundamentals and database development technologies. • Prepare datasets for the application of given analytics or data mining techniques. • Apply data analytics or data mining skills to given datasets and evaluate the results obtained. Postgraduate Diploma

Educational Aims of the Course	To produce practitioners who, in addition to the aims above, will be able to: • Demonstrate knowledge and understanding of information retrieval systems. • Critically appraise characteristics and capabilities of modern data stores. • Design, implement and evaluate advanced data analytics and retrieval systems applied to Big Data. • Establish a data management framework and lay down effective procedures for data warehousing. • Demonstrate knowledge and understanding of the techniques required to manage and use distributed, unstructured data stores. • Critically appraise state of the art techniques in data science. Master of Science In addition to the above, MSc graduates will be able to: • Analyse the requirements of a substantial data science problem. • Critically appraise different design approaches and technical solutions for a data science project. • Plan, manage and execute a substantial data science project. • Integrate different data science technologies. • Effectively test, evaluate and document a complex data science solution.
Awarding Institution	Robert Gordon University
Course Duration	Full-Time Students Full-time 12 months with a September intake Full-time 16 months with a January intake Part-Time Students Part-time 28 months with a September intake Part-time 32 months with a January intake Depending on their mode of study, students can choose to undertake either a short-term (3 months) or long-term (12 months) placement. All placements require approval from the School prior to commencement. If a student goes on placement, the duration of the course will be extended by the duration of the placement.
Distinctive Features of the Course	The course offers specialist advanced technical skills in the areas of Modern Database Systems, Data Mining and Warehousing, Information Retrieval and Extraction, Big Data Analytics and Visualisation, Data Science Development and application of Advanced Data Science techniques. The course will benefit from collaborations with the National Subsea Centre, the Data Lab and industry; the course will also gain from inter-university links with other Scottish universities.

	The long placement of up to 12 months, offers students the opportunity to develop computing skills and build teamwork and networking skills in a realistic business environment. There is also a 3 month short placement opportunity to develop the student's professional and technical skills in a business environment.
Placement/Professional Experience	Students may elect to undertake a short placement or a year-long placement. The placement provides an opportunity to sample working life, be it in commerce, industry or the public sector, and apply knowledge and skills learned in the classroom to the challenges of the workplace. Students who have completed the placement as part of their course have a definite advantage when it comes to applying for graduate positions. With the combination of training, experience, and a successful degree, students are often offered subsequent employment by the placement provider with which they undertook their placement. While the student is on placement, they will be responsible to a designated member of the placement provider (the Industrial Supervisor) for the discharge of their day-to-day duties. Each student on placement is also assigned an Academic Tutor from the staff of the School, who will act as the student's personal tutor during the placement. The Academic Tutor will hold a small number of scheduled video calls with the student and their Industrial Supervisor during the placement. Students may choose to undertake a short placement. This normally comprises 3 months of continuous, full-time supervised work experience. The assessment for a short placement experience is based on a review of the student's work-related skills, conducted by the Industrial Supervisor. Students on a short placement will normally receive at least 1 video call with their Academic Tutor. As an alternative, students may undertake a year-long placement. The year-long industrial placement normally comprises of 12 months of continuous, full-time supervised work experience. The assessment for a year-long placement is based on a review of the student's work-related skills, conducted by both the student and their Industrial Supervisor. Students on a year-long placement will normally receive at least 2 video calls with their Academic Tutor. Placements must be undertaken after the completion of all taught modules and begin before the start of the final project. A student may choose to either undertake a placement at the same time as their final project or delay the start of their final project until the placement has finished. The course leader must approve this decision. The University provides support to students interested in undertaking a placement through the Placement Office and Centre for Employability and Community Engagement. The Placement Office maintains a pool of placement opportunities and establishes contact with new placement providers as necessary. It will normally assist students by informing them of suitable placement opportunities. Students are also given help and guidance by the Placement Office in obtaining their own placement from those advertised directly by companies or through their own connections. The Centre for Employability and Community Engagement offers further support for students, including with preparing their CV and developing interview skills. All placements require approval from the School prior to commencement, in order to ensure that they are suitable for the student and to confirm that the placement provider has put adequate supervisory and health-and-safety arrangements in place. Placements cannot be guaranteed and students wishing to undertake a placement are expected to reasonably engage with the process, for example by providing a copy of their CV to the Placement Office and by attending any interviews with potential placement providers when required. The credits awarded for the placement module are in addition to the 180 credits achievable from the other modules. Successful completion is not required for progression or graduation. The credits achieved for placement may not be used to compensate for any credits from the other modules.

Opportunities for Further Study	The course has a strong grounding in research techniques and enables students to advance into research and development at doctoral level within this institution or other UK higher education institutions.
External and Internal Indicators of Quality and Standards	SCQF credit values have been used as a guide during the design of modules while the structure and content of the course has also been designed with the British Computer Society's (BCS) requirements for Practical/Engineering Applications in mind, and embedded in the core structure are modules and topics which provide the student with computing skills and knowledge in Underlying Theory, Design and Methodology, Tools and Techniques and Project Management and Group work. Standards are primarily established through QAA (Quality Assurance Agency) Benchmarks and the requirements imposed by the British Computer Society in order to gain professional course recognition. The achievement of appropriate standards is demonstrated through the University's course validation procedures and professional institution accreditation visits. The QAA qualification descriptors and Subject Benchmark Statements provide a reference framework that has guided the evolution of the School's programme, and the extent of the subject discipline, as defined by QAA, is broadly matched by the provision that now exists within the School's programmes. A number of internal processes and procedures are used by the School to ensure that standards are maintained and enhanced. Stakeholders' views on standards are routinely obtained from the Staff/Student Liaison Committee, Programme Management Teams, Student Experience questionnaires, Industrial Board, industrial visits to placement students, staff involvement in professional body activities and staff participation in accreditation events outside the School and at other HE institutions. In addition to this, best practice in teaching and learning within the University is monitored through the University quality enhancement and quality assurance processes. These include the regular five-yearly cycle of Institution-Led Subject Reviews, as well as participation in the University quality assurance committee structures, such as the University's Student Experience, and Academic Regulations Sub-Committees. Finally, it should be mentioned that the lecturers within the School have an ongoing interest in pedagogical research and a commitment to expanding the scholarship of computing science education. Academic staff help to organise, and participate in, national and international conferences and have been awarded funding from the Higher Education Academy, the Scottish Informatics and Computer Science Alliance, and the Scottish Enhancement Themes to pursue work in computer science education, as well as more general work on student transitions. The School also participates in university conferences on teaching and learning.
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.

Course-Specific Entry Requirements	The normal minimum entry requirements are an Honours Degree (2nd class or better) in a discipline with significant Computing, Mathematics or Statistics content.
Alternative Qualifications for Entry	Applicants with a lesser qualification who possess significant industrial experience or substantial relevant personal development in a relevant field may be considered for entry into the course.
Interview Procedures	Where an applicant supports an application with alternative qualifications, the School may wish to arrange an interview to discuss the level of qualification offered and the experience gained in an industrial background.
Course-Specific Regulations	The MSc Project module cannot be condoned as it is key to demonstrating essential skills in the course such as the ability to conduct a significant piece of independent work. Furthermore, BCS guidelines state that the 60 credit project should not be compensated. The placement modules cannot be used to compensate for failed modules. All students, regardless of mode and start date, will typically undertake resits in the summer exam period. Exceptionally students who start in January may be offered a resit opportunity after completing two semesters, in April of the following year.
Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
	The Purpose of the University is: - To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: - To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability.

University Strategy	The Culture of the University is: • To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK. RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.
	The extent of the course syllabus demands a relatively rapid pace of delivery. Consequently, student-centred learning strategies play an important role in most subjects and throughout all stages, reinforcing and supplementing more traditional methods of subject delivery. Directed reading, independent progress through computer-based exercises, independent reading, analysis of case studies and directed searches of on-line information resources are all commonplace. The preferred strategy for any course module depends on the nature and extent of the syllabus content, on its subject-specific learning outcomes and its contribution to the overall course aims. In the sections that follow, a more detailed description of the learning experience is given, which distinguishes between: • Core technological modules • Professional skills modules • Integrative course modules Teaching and learning are reviewed through an annual appraisal mechanism and strategies are liable to change for the sake of continuing improvement and to reflect best practice. Part-time students participate in the same sessions as full-time students, but they take 2 modules (30 credits) per semester instead of 4 modules (60 credits). <u>Core Technological Modules</u>

Course-Specific Learning Experience	These modules impart specialist technical knowledge through the medium of lectures and tutorials, supplemented as necessary by student-centred materials and tasks. All core modules have a supporting practical element, which is delivered through laboratory work. In practical sessions students are expected to progress through a series of graded exercises that are designed to foster analytical skills and to illustrate the synthesis of fundamental knowledge and problem solving techniques in a structured problem domain. For DL students, the traditional lecture will be replaced by on-line materials together with on-line support sessions. <u>Professional Skills Modules</u> Technical report writing skills are exercised in all modules. The course addresses project planning skills that are vocationally relevant and that also directly underpin the planning and management of the MSc Project. Legal issues are learnt via group work tasks. Student feedback in this course takes the form of critical self-appraisal, peer appraisal, and commentary from academic staff. Through directed reading and by listening to invited external speakers, students are also encouraged to develop a broader perspective of Data Science and to sharpen their awareness of the need for professional standards and ethics in the workplace. Recorded seminars will be made available to DL students where possible. <u>Integrative Course Modules</u> The MSc Project modules serve the primary purpose of integrating technological and professional strands, which are developed in the preceding PgD stage, and typically do so in the context of a substantial data science project. The project module typically involves the research, planning, development and evaluation of the solution to a problem, which occurs within a relatively unstructured domain. The problem is original to the student and its solution therefore requires the innovative application of knowledge and techniques either studied in the previous PgD stage or acquired through independent research of recent and relevant literature. The MSc Project provides a vehicle for integrating specialist knowledge with analytic, problem solving, managerial and communication skills. All of these are exercised and evidenced through the execution and outcomes of the project. <u>Research Skills</u> The course framework underpins the MSc Project in a number of ways: PgC / PgD taught modules provide support in specific subject areas, designed to foster understanding and competence in project management and planning issues and in the testing and evaluation of implemented systems. Because of the novel nature of the project and the rapid evolution of computing, students also require competence in research skills. These skills are intended to assist the student in the investigation of emerging techniques and technologies that may be relevant to the achievement of the project goals. (Beyond the confines of the MSc Project, these same research skills are useful for continuing professional development and act as a safeguard against technological obsolescence.) <u>Group Work</u> Group exercises are used to develop team-working skills and encourage discussion. This type of work will be used in some technological core modules as well as in the Professional Skills for Data Science module, where students will investigate legal issues with respect to one or more cases. DL students will participate in group exercises wherever possible, but the communication of members of the group will be different. For example, chat may be used instead of a face to face discussion and documents may be emailed instead of given in person.
	The course curriculum is designed to enable the student to acquire and develop their subject knowledge and understanding, and to gain practical and key transferable skills at Masters level. Specifically to practice and apply their knowledge skills and understanding, critically apply cognitive skills, develop and practice a wide range of communications skills, and

Teaching and Learning Strategy	exercise autonomy and accountability when working with others, consistent with the corresponding Scottish Credit and Qualifications Framework level 11 guidelines. As the students progress through the course they are encouraged to expand their understanding and critical appreciation of advanced computing, and develop skills in programming through techniques and tools with the application of this knowledge. The overall philosophy and strategy for the course is to provide a sound knowledge and understanding of advanced computing in the broader context. The vocationally orientated nature of the course requires the student to achieve general and specific skills, which are addressed through a combination of teaching and learning methods, including lectures, lab sessions, tutorials, seminars, student-centred learning and private study (directed reading of textbooks and online material) and structured laboratory sessions. The broad nature of the programme including common and specialist elements necessitates the use of a broad range of teaching techniques. These include lectures, seminars, tutorials, computer laboratories, open and distance learning and group work on a large project. Directed guest lectures are used to encourage and emphasise professional skills. Course induction and directed study promote development and enhancement of skills in subject context throughout course. A handbook on basic study skills is provided to all students. Learning skills are developed through tutorials, directed reading and projects. Tutorials, case studies and graded problem assignments e.g. in laboratory sessions are employed to provide practical and theoretical underpinning of subject matter provided by lectures. Students meet regularly with project supervisors during MSc project modules.
Assessment	A range of assessment methods are used, with particular emphasis placed on coursework assignments, including design exercises (ranging from design studies to fully implemented solutions), comparative studies, and investigative reports. Students are told in advance what assessment methods will be used for each module and will have access to examples of past assessments. Where the format of assessment varies significantly from one year to the next, examples of assessment will be created in order to ensure students understand what the assessment entails. All assessments include a marking/grading scheme which describes the criteria used for the award of marks/grades. Structured project documents will specify the management of the project conduct, quality guidelines and project deliverables. It will also contain information on the use of web resources to present interim MSc Project progress updates and guidelines on the live presentation and demonstration of MSc Project outcomes. The use of IT tools will be assessed in all coursework assignments (i.e. use of tools to plan, aid design, collate results and document coursework). Demonstration of IT competence is mandatory in production of documentation to support all coursework submissions.
University Support for Teaching, Learning and Assessment	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • a Team Around a Student (TAAS) system of pastoral support to students including online support www.rgu.ac.uk/policies [The TAAS School Student Support Model replaces the Personal Tutor model of student support from Session 2026-27]; • an extensive programme of student study skills delivered through Student Life and the Library; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in Scotland's Tertiary Enhancement Programme (STEP) www.step.ac.uk ; • support for staff from the Department of Academic Quality and Learning Enhancement (AQLE), including credit-rated teaching, learning and assessment provision;

	• extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • Moodle, http://campusmoodle.rgu.ac.uk , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
Course-Specific Support for Teaching, Learning and Assessment	The School's computing facilities comprise 9 labs in total housed with over 335 state-of-the-art PCs and 30 Macs, each updated over a 3 year cycle to ensure they deliver high performance and support the latest software. School facilities include the School's Mac Suite, a soundproof booth used for podcasting, a Green Screen Suite used in multimedia video production and a high performance computer cluster used to create computer generated imagery. The School offers 100% wireless network coverage for laptops and other handheld devices. There's also a comprehensive range of software on-site - some developed by staff to support coursework and making wide use of open source and public domain technologies. Established relationships with major software suppliers mean access to the latest software. Access to labs is by electronic key fob at any time during University opening hours. Students have access to the University's wireless network for laptops and other handheld devices. All students are given access to the Student handbook, which contains a wide variety of information about the School and University. All students have a personal tutor who is responsible for their pastoral care. The weekly induction sessions during the first semester of study cover a wide variety of topics such as library facilities, extenuating circumstances, exam preparation, careers services, placement opportunities, etc. All students are assigned a supervisor for their project work. In addition, they receive extensive guidelines on how to undertake a project. The School believes the academic success of the students is founded in the desire to develop and maintain close links with industry and to actively seek state-of the-art technologies and integrate these into course modules.
University Monitoring of Quality and Standards	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; • <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams;

	• School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	The School is committed to the highest standards of quality and best pedagogy and research within an environment which values all associated stakeholders. Formal feedback is gathered from Student Questionnaires which are issued on completion of the PgDip stage of the course. It offers the students the opportunity to comment in detail on all aspects of the course as well as the general environment, support services and facilities. Responses to, and actions resulting from, these comments are addressed fully in an Annual Course /Programme Appraisal and widely discussed by the appropriate Programme Management Team. The Course Team also runs a series of informal focus groups in order to gather both positive and negative comments on the course. Where problems are highlighted, these are discussed fully and potential solutions are considered. Less formally, the students have an opportunity to discuss with their Course Leader any course or management issues.
University Student Feedback Mechanisms	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements. This approach is underpinned by the <i>Student Partnership Agreement</i> which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: Student Representation & Partnership RGU . There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience. Mechanisms through which students are engaged and supported include: • participation in student evaluation questionnaires and where relevant the <i>National Student Survey (NSS)</i> ; • staff/student liaison arrangements at course/programme level; • support for Student Representatives; • student representation on Institution-Led Subject Review Panels and Validations; • student representation on key University committees.
	The University uses a variety of mechanisms to obtain feedback from students about their learning experience. The University has a system of student representation, via the appointment of Student Representatives for each course, who in turn represent the interests of their colleagues at Staff/Student Liaison meetings held each semester with staff representatives from the School concerned. Student Representatives are also invited to attend a meeting with the Dean of School during the first and second semester to discuss a

Course-Specific Student Feedback Mechanisms	selection of topics and to air any concerns they have. The notes of all these meetings and responses to any issues raised are subsequently made available. A single Staff/Student Liaison Committee exists for all postgraduate courses in the School, which meets regularly. The Committee comprises the Dean of School, Course Leaders, teaching staff and class representatives from each course intake and meets at least once per semester to exchange information and discuss issues.
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Awards
Stage 1

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Postgraduate Certificate - PgCert	Data Science	Aberdeen, United Kingdom	Exit Award		60
Postgraduate Diploma - PgDip	Data Science	Aberdeen, United Kingdom	Exit Award		120
Master of Science - MSc	Data Science	Aberdeen, United Kingdom	Target Award		180

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	Demonstrate knowledge and understanding of modern NoSQL and relational database systems, and advanced data mining methods.	SC1	SCQF 11
LLO2	Demonstrate knowledge and understanding of tools and techniques which aid the specification, design and implementation of a data science project.	SC1	SCQF 11
LLO3	Demonstrate knowledge and understanding of knowledge based techniques which can be applied to analyse and extract useful information from complex data stores.	SC1	SCQF 11
LLO4	Demonstrate knowledge and understanding of how to select and establish a project management framework and lay down effective procedures for monitoring progress of any large scale research/development project.	SC1	SCQF 11
LLO5	Demonstrate knowledge and understanding of the availability of information resources and how to access these to obtain state-of-the-art knowledge of current data science and Big Data technologies.	SC1	SCQF 11
LLO6	Demonstrate knowledge and understanding of comprehensive, detailed, state-of-the-art knowledge of the specialist area(s) covered by the course within the context of the broader discipline or field within which the course is set.	SC1	SCQF 11
LLO7	Demonstrate knowledge and understanding of research methods appropriate specifically to their Masters independent study project, together with detailed knowledge of the particular area or sub-area in which the project is carried out.	SC1	SCQF 11
LLO8	Demonstrate knowledge and understanding of the theoretical and empirical limits and boundaries of data science, and the range of methods of study and types of judgments employed by advanced practitioners.	SC1	SCQF 11
LLO9	Demonstrate the skills required to apply statistical methods to assess and evaluate research or experimental test results.	SC2	SCQF 11
LLO10	Demonstrate the skills required to use a wide range of data science tools, techniques and skills.	SC2	SCQF 11
LLO11	Demonstrate the skills required to make effective use of available IT tools and packages to undertake the practical work of the PgC/PgD modules, to produce high quality coursework.	SC2	SCQF 11
LLO12	Demonstrate the skills required to make effective use of available IT tools and packages to undertake a data science project and to enhance the production and presentation of the Masters stage project report and oral defence.	SC2	SCQF 11
LLO13	Demonstrate the ability to critically analyse a data science problem identifying key requirements, alternative solutions and evaluation methods.	SC3	SCQF 11
LLO14	Demonstrate the ability to develop an effective, original solution to a data science problem.	SC3	SCQF 11
LLO15	Demonstrate the ability to make calculated decisions in situations where information is incorrect and/or incomplete.	SC3	SCQF 11
LLO16	Demonstrate the ability to propose solutions to data storage, retrieval, extraction, analysis and visualisation problems which satisfy user requirements, using appropriate notations to express solution concepts and aid reasoning about the efficacy of the proposed design.	SC3	SCQF 11
LLO17	Demonstrate the ability to evaluate the efficacy of proposed solutions.	SC3	SCQF 11
LLO18	Demonstrate the ability to apply critical judgment in the selection and application of analysis / test strategies for validating research outcomes / testing a developed system and compliance to initial requirements.	SC3	SCQF 11
LLO19	Demonstrate the ability to propose (or select) and conduct the Masters project applying an analytic, rigorous and critical approach to identifying project goals, proposing and evaluating different research/solution strategies and generating a useful and effective output.	SC3	SCQF 11
	Demonstrate the skills required to communicate effectively, using appropriate methods to audiences with different levels of knowledge and		

LLO20	understanding.	SC4	SCQF 11
LLO21	Demonstrate the skills required to produce structured and coherent written reports to document the investigation and survey of a problem domain or a data science task.	SC4	SCQF 11
LLO22	Demonstrate the skills required to formulate strategy / design decisions and record research / test strategies and results throughout the various stages of a major data science project using appropriate software tools.	SC4	SCQF 11
LLO23	Demonstrate the skills required to produce a clear and coherent thesis which documents the process and end result of a major research / data science exercise.	SC4	SCQF 11
LLO24	Demonstrate the skills required to analyse a variety of data (including numerical data).	SC4	SCQF 11
LLO25	Demonstrate the skills required to deliver a lucid and coherent oral defence of the work undertaken during the Masters project before a live audience of peers and/or academics.	SC4	SCQF 11
LLO26	Demonstrate the skills required to deliver concise and relevant reviews of work undertaken within a predefined plan of work and make a valuable contribution to team work tasks.	SC4	SCQF 11
LLO27	Demonstrate the ability to investigate and critically appraise specialist areas of research.	SC5	SCQF 11
LLO28	Demonstrate the ability to use tools and techniques to analyse existing complex data.	SC5	SCQF 11
LLO29	Demonstrate the ability to work as part of a team, contributing effectively and taking significant responsibility for the work of the team.	SC5	SCQF 11
LLO30	Demonstrate the ability to work independently, demonstrating a high level of autonomy and own initiative, taking responsibility for own work.	SC5	SCQF 11
LLO31	Demonstrate the ability to communicate effectively, both orally and in writing.	SC5	SCQF 11
LLO32	Demonstrate the ability to integrate critical reflection into work roles and responsibilities; adopt an analytical approach to tasks to identify problems, solutions and evaluation methods.	SC5	SCQF 11
LLO33	Demonstrate the ability to show a willingness to learn from others and problem solve.	SC5	SCQF 11
LLO34	Demonstrate the ability to self-manage time and workload scheduling flexibly.	SC5	SCQF 11
LLO35	Demonstrate the ability to consider the ethical, social and professional issues of a data science problem/solution, effectively addressing any such issues.	SC5	SCQF 11
LLO36	Demonstrate the ability to critically analyse problems and solution alternatives.	SC5	SCQF 11

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
		SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9

SC2	Practice: Applied knowledge, skills and understanding	SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC5	Autonomy, accountability and working with others	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

Delivery Range

<i>Title</i>	<i>Description</i>
Full Time On Campus	Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.
Part Time On Campus	Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.
Part Time Online	Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.

Stage 1 / Semester 1**Full Time On Campus**

Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.

Part Time On Campus

Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.

Part Time Online

Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Part Time Online	CMM524	Advanced Data Management		Kit-ying Hui (cmskhu)	Yes	2.0	15	Level 11
Full Time On Campus	CMM524	Advanced Data Management		Kit-ying Hui (cmskhu)	Yes	2.0	15	Level 11
Part Time On Campus	CMM524	Advanced Data Management		Kit-ying Hui (cmskhu)	Yes	2.0	15	Level 11
Full Time On Campus	CMM510	Data Mining		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
Part								

Time Online	CMM535	Data Science Development		David Lonie (dpl0795)	Yes	2.0	15	Level 11
Full Time On Campus	CMM535	Data Science Development		David Lonie (dpl0795)	Yes	2.0	15	Level 11
Part Time On Campus	CMM535	Data Science Development		David Lonie (dpl0795)	Yes	2.0	15	Level 11
Full Time On Campus	CMM531	Data Warehousing		Mark Snaith (ms10541)	Yes	2.0	15	Level 11

Stage 1 / Semester 2

Full Time On Campus

Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.

Part Time On Campus

Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.

Part Time Online

Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.

Core								
Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
Full Time On Campus	CMM536	Advanced Data Science		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
Full Time On Campus	CMM534	Big Data Analytics and Visualisation		David Lonie (dpl0795)	Yes	2.0	15	Level 11
Full Time On Campus	CMM544	Natural Language Processing		Laura Tucker	Yes	2.0	15	Level 11
Part								

Time On Campus	CMM544	Natural Language Processing		Laura Tucker	Yes	2.0	15	Level 11
Part Time Online	CMM544	Natural Language Processing		Laura Tucker	Yes	2.0	15	Level 11
Part Time Online	CMM507	Professional Development and Research Skills		Verda Munir	Yes	2.0	15	Level 11
Full Time On Campus	CMM507	Professional Development and Research Skills		Verda Munir	Yes	2.0	15	Level 11
Part Time On Campus	CMM507	Professional Development and Research Skills		Verda Munir	Yes	2.0	15	Level 11

Stage 1 / Semester 3**Full Time On Campus**

Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.

Part Time On Campus

Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.

Part Time Online

Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Part Time On Campus	CMM510	Data Mining		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
Part Time Online	CMM510	Data Mining		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
Part Time Online	CMM531	Data Warehousing		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
Part Time On Campus	CMM531	Data Warehousing		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
Full Time On Campus	CEM100	MSc Project	Routes A, B, C	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
Full Time				Mark Bartlett				Level

On Campus	CEM105	Year-long Postgraduate Placement	Route C	(mb9400)	No	1.0	120	11
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Stage 1 / Semester 4

Part Time On Campus

Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.

Part Time Online

Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.

Core								
Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
Part Time Online	CMM536	Advanced Data Science		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
Part Time On Campus	CMM536	Advanced Data Science		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
Part Time Online	CMM534	Big Data Analytics and Visualisation		David Lonie (dpl0795)	Yes	2.0	15	Level 11
Part Time On Campus	CMM534	Big Data Analytics and Visualisation		David Lonie (dpl0795)	Yes	2.0	15	Level 11

Stage 1 / Semester 5**Part Time On Campus**

Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.

Part Time Online

Route A - Non-Placement. Route B - Short Placement. Route C - Long Placement.

Core								
Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
Part Time On Campus	CEM100	MSc Project	Routes A, B, C. Undertaken over two semesters.	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
Part Time Online	CEM100	MSc Project	Routes A, B, C. Undertaken over two semesters.	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
Elective								
Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
Part Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
Part Time Online	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
Part Time On Campus	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11
Part Time Online	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Course Deliveries

<i>Stage</i>	<i>Semester</i>	<i>Delivery Type</i>	<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
1	Semester 1	Core	Part Time Online	CMM524	Advanced Data Management		Kit-ying Hui (cmskhu)	Yes	2.0	15	Level 11

1	Semester 1	Core	Full Time On Campus	CMM524	Advanced Data Management		Kit-ying Hui (cmskhu)	Yes	2.0	15	Level 11
1	Semester 1	Core	Part Time On Campus	CMM524	Advanced Data Management		Kit-ying Hui (cmskhu)	Yes	2.0	15	Level 11
1	Semester 1	Core	Full Time On Campus	CMM510	Data Mining		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
1	Semester 1	Core	Part Time Online	CMM535	Data Science Development		David Lonie (dpl0795)	Yes	2.0	15	Level 11
1	Semester 1	Core	Full Time On Campus	CMM535	Data Science Development		David Lonie (dpl0795)	Yes	2.0	15	Level 11
1	Semester 1	Core	Part Time On Campus	CMM535	Data Science Development		David Lonie (dpl0795)	Yes	2.0	15	Level 11
1	Semester 1	Core	Full Time On Campus	CMM531	Data Warehousing		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	CMM536	Advanced Data Science		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	CMM534	Big Data Analytics and Visualisation		David Lonie (dpl0795)	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	CMM544	Natural Language Processing		Laura Tucker	Yes	2.0	15	Level 11

			Campus								
1	Semester 2	Core	Part Time On Campus	CMM544	Natural Language Processing		Laura Tucker	Yes	2.0	15	Level 11
1	Semester 2	Core	Part Time Online	CMM544	Natural Language Processing		Laura Tucker	Yes	2.0	15	Level 11
1	Semester 2	Core	Part Time Online	CMM507	Professional Development and Research Skills		Verda Munir	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	CMM507	Professional Development and Research Skills		Verda Munir	Yes	2.0	15	Level 11
1	Semester 2	Core	Part Time On Campus	CMM507	Professional Development and Research Skills		Verda Munir	Yes	2.0	15	Level 11
1	Semester 3	Core	Part Time On Campus	CMM510	Data Mining		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
1	Semester 3	Core	Part Time Online	CMM510	Data Mining		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
1	Semester 3	Core	Part Time Online	CMM531	Data Warehousing		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
1	Semester 3	Core	Part Time On Campus	CMM531	Data Warehousing		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
1	Semester 3	Core	Full Time On Campus	CEM100	MSc Project	Routes A, B, C	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
			Full								

1	Semester 3	Elective	Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
1	Semester 3	Elective	Full Time On Campus	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11
1	Semester 4	Core	Part Time Online	CMM536	Advanced Data Science		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
1	Semester 4	Core	Part Time On Campus	CMM536	Advanced Data Science		Mark Snaith (ms10541)	Yes	2.0	15	Level 11
1	Semester 4	Core	Part Time Online	CMM534	Big Data Analytics and Visualisation		David Lonie (dpl0795)	Yes	2.0	15	Level 11
1	Semester 4	Core	Part Time On Campus	CMM534	Big Data Analytics and Visualisation		David Lonie (dpl0795)	Yes	2.0	15	Level 11
1	Semester 5	Core	Part Time On Campus	CEM100	MSc Project	Routes A, B, C. Undertaken over two semesters.	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
1	Semester 5	Core	Part Time Online	CEM100	MSc Project	Routes A, B, C. Undertaken over two semesters.	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
1	Semester 5	Elective	Part Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
1	Semester 5	Elective	Part Time Online	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
1	Semester 5	Elective	Part Time On	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11

			Campus								
1	Semester 5	Elective	Part Time Online	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
Aberdeen, United Kingdom	On-Campus	Full Time	
Aberdeen, United Kingdom	On-Campus	Part Time	
All Applicable Locations	Online	Part Time	

Sources of Further Information

No sources of further information recorded
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